
Metal-Sci: A Scientific Compute Benchmark for Evolutionary LLM Kernel Search on Apple Silicon

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Abstract

We present METAL-SCI, a 10-task benchmark of scientific Apple Silicon Metal compute kernels spanning six optimisation regimes (stencils, all-pairs, multi-field Boltzmann, neighbour-list MD, multi-kernel PDE, FFT). Each task ships a CPU reference, a roofline-anchored fitness, and a held-out generalisation size. We report matched single-model sweeps of Claude Opus 4.7 and Gemini 3.1 Pro Preview on M1 Pro: in-distribution self-speedups span $1.00\times$ to $10.7\times$, and the held-out gate exposes specialisation/generalisation tradeoffs the geometric mean hides — notably an Opus template `<uint D> HMC` win that returns wrong answers at unseen $d=24$ where Gemini’s variant generalises at $17.8\times$.

1 Introduction

LLM code-search systems — FunSearch [1], AlphaEvolve [2], and recent kernel-generation work [3] — evaluate language models as *optimisers* of executable artefacts. The choice of artefact matters: KernelBench targets PyTorch ML kernels (GEMM, attention, convolution) on CUDA, where canonical implementations are heavily represented in pretraining data and the optimisation surface is dominated by tiling and warp-level reductions.

Scientific computing exposes a different surface. Stencils are bandwidth-bound and reward halo and temporal blocking. All-pairs interactions are compute-bound and reward register blocking with shared-memory cooperative loads. Lattice Boltzmann ping-pongs nine distribution functions per cell with non-trivial pull-stream indexing. Cell-list molecular dynamics touches irregular memory through atomic counters. Hamiltonian Monte Carlo runs many chains in parallel where register pressure scales with the problem dimension d . Grad-Shafranov solvers chain a max-reduction with a variable-coefficient stencil. 3D FFTs are dominated by data-shuffle/butterfly patterns and twiddle-factor caching rather than tiling. Each pattern stresses a distinct dimension of the compiler/memory hierarchy, and canonical optimisations [5,6,7,8,11] differ regime-to-regime.

We target Apple Silicon’s Metal compute pipeline. It is underrepresented in CUDA-centric training data — frontier models reliably mishandle Metal-specific syntax such as the `[[max_total_threads_per_threadgroup]]` kernel attribute, a real out-of-distribution generalisation test. Its unified memory model removes host-device copy plumbing, enabling sub-second compile-run-verify cycles per candidate that are essential for fast evolutionary loops. And it supports rich simdgroup intrinsics (`simd_max`, `simd_broadcast`) and threadgroup-memory tiling, giving the LLM a non-trivial optimisation surface.

Contributions. (i) A 10-task scientific compute benchmark over six optimisation regimes, each with a CPU reference, roofline-anchored fitness, and multi-size generalisation gate. (ii) A runtime-compiled harness that exposes compile errors, correctness diagnostics, and per-size GPU timing back to the LLM. (iii) Two matched single-model sweeps — Claude Opus 4.7 and Gemini 3.1 Pro Preview — on M1 Pro, with in-distribution *and* held-out evaluation per task; the held-out gate

exposes specialisation/generalisation tradeoffs the in-distribution geometric mean hides, and the two models overfit asymmetrically.

2 Benchmark tasks

Each task ships (a) a Metal source describing the problem, (b) a naive correct seed, (c) a CPU reference with task-specific tolerance, (d) three in-distribution sizes plus one held-out size, and (e) a roofline ceiling in either GFLOPS (compute-bound) or GB/s (bandwidth-bound). Tasks are grouped by optimisation regime.

2.1 Regular stencils (regime R1)

heat2d. Two-dimensional heat equation, 5-point stencil on $(N_x \times N_y)$ grid with Dirichlet boundaries. Per timestep, per interior cell:

$$u_{i,j}^{n+1} = u_{i,j}^n + \alpha (u_{i-1,j}^n + u_{i+1,j}^n + u_{i,j-1}^n + u_{i,j+1}^n - 4u_{i,j}^n). \quad (1)$$

Bandwidth-bound at 8 B/cell; tests halo handling and temporal blocking.

wave3d. Three-dimensional acoustic wave equation, 7-point isotropic Laplacian, leapfrog in time:

$$u_{i,j,k}^{n+1} = 2u_{i,j,k}^n - u_{i,j,k}^{n-1} + \alpha (u_{i\pm 1,j,k}^n + u_{i,j\pm 1,k}^n + u_{i,j,k\pm 1}^n - 6u_{i,j,k}^n), \quad (2)$$

where each $\pm$ expands to two terms. CFL stability requires $\alpha = (c\Delta t/\Delta x)^2 < 1/3$ (we use 0.18). 12 B/cell unique DRAM traffic. Tests register pressure, 2.5D blocking, and marching-axis choice.

2.2 Compute-bound (regime R2)

nbody. All-pairs gravitational N -body with leapfrog. Per body i per step:

$$\mathbf{a}_i = G \sum_j m_j \frac{\mathbf{r}_j - \mathbf{r}_i}{(\|\mathbf{r}_j - \mathbf{r}_i\|^2 + \varepsilon^2)^{3/2}}, \quad \mathbf{v} += \mathbf{a}\Delta t, \quad \mathbf{r} += \mathbf{v}\Delta t. \quad (3)$$

Self-interaction is masked by the softening ε . ~ 20 FLOPs per pair; ceiling at peak FP32 GFLOPS. Tests register blocking and threadgroup cooperative loads.

hmc. Hamiltonian Monte Carlo on an anisotropic Gaussian target, $U(q) = \frac{1}{2}q^\top Aq$ with $A = \Sigma^{-1}$. One thread per chain; many chains in parallel. Each HMC step draws $p \sim \mathcal{N}(0, I)$ and runs L leapfrog inner steps with stepsize ε ,

$$p \leftarrow p - \frac{\varepsilon}{2} Aq, \quad q \leftarrow q + \varepsilon p, \quad p \leftarrow p - \varepsilon Aq, \dots, \quad p \leftarrow p - \frac{\varepsilon}{2} Aq, \quad (4)$$

followed by a Metropolis accept/reject with $\log u_{\text{acc}} < -\Delta H$. Correctness is verified statistically (sample mean and Frobenius covariance error vs. the target). Three sizes $(d, K) \in \{(8, 16K), (16, 4K), (32, 1K)\}$ probe the register-pressure boundary; at $d=32$ per-thread state (~ 512 B) competes with the register file.

2.3 Multi-field, exotic memory (regime R3)

lbm. D2Q9 Lattice Boltzmann, fused pull-stream + BGK collision, periodic BC. With velocity table $\mathbf{c}_k$ and weights w_k , per cell per step:

$$f_k^{\text{str}}(\mathbf{x}) = f_k^{\text{in}}(\mathbf{x} - \mathbf{c}_k), \quad \rho = \sum_k f_k^{\text{str}}, \quad \mathbf{u} = \frac{1}{\rho} \sum_k \mathbf{c}_k f_k^{\text{str}}, \quad (5)$$

$$f_k^{\text{eq}} = w_k \rho \left(1 + 3(\mathbf{c}_k \cdot \mathbf{u}) + \frac{9}{2}(\mathbf{c}_k \cdot \mathbf{u})^2 - \frac{3}{2}\|\mathbf{u}\|^2 \right), \quad (6)$$

$$f_k^{\text{out}} = f_k^{\text{str}} - \tau^{-1}(f_k^{\text{str}} - f_k^{\text{eq}}). \quad (7)$$

Storage is SoA: $f[k N_x N_y + j N_x + i]$. 72 B/cell DRAM traffic. Tests AoS/SoA layout, push vs pull streaming, and algebraic factorisations of the BGK kernel.

ising. Two-dimensional Ising model, checkerboard Metropolis Monte Carlo on a periodic $N_x \times N_y$ lattice with $\sigma \in \{\pm 1\}$ stored as int8. Per attempt at site (i, j) :

$$h = \sigma_{i\pm 1, j} + \sigma_{i, j\pm 1} \in \{-4, \dots, 4\}, \quad \Delta E = 2J\sigma h, \quad \text{accept} \iff u < \exp(-\beta \Delta E). \quad (8)$$

A precomputed five-entry p_{accept} table and a counter-based Murmur-fmix32 PRNG yield bit-exact CPU/GPU agreement: verification is byte-equality on the spin array. 2 B/site/sweep ceiling.

2.4 Irregular memory and atomics (regime R4)

lj. Lennard-Jones molecular dynamics with a cell-list spatial hash. Three kernels per step — `clear_cells`, `build_cells` (atomic scatter), `step` (27-neighbour-cell pair force + integration):

$$\mathbf{F}_i = \sum_{j \in \mathcal{N}(i)} -24(2r_{ij}^{-12} - r_{ij}^{-6}) r_{ij}^{-2} (\mathbf{r}_j - \mathbf{r}_i), \quad r_{ij} < r_{\text{cut}} = 2.5, \quad (9)$$

with minimum-image periodic wrap. Cell occupancy is built via `atomic_fetch_add` on a per-cell counter. Tests load balancing, atomic contention, and the irregular access patterns of neighbour-list MD.

2.5 Multi-kernel reductions (regime R5)

gradshaf. Grad-Shafranov fixed-boundary equilibrium via Picard iteration, two kernels per outer step: an interior max-reduction $\psi_{\text{axis}} = \max_{(i,j) \in \text{int}} \psi$, then a variable-coefficient 5-point stencil with nonlinear source:

$$\psi_{\text{norm}} = \psi / \psi_{\text{axis}}, \quad J = R p_{\text{axis}} \cdot 4\psi_{\text{norm}}(1 - \psi_{\text{norm}}) \cdot \mathbf{1}[0 < \psi_{\text{norm}} < 1], \quad (10)$$

$$\Delta^* \psi = a_W \psi_W + a_E \psi_E + a_N \psi_N + a_S \psi_S + a_C \psi_C, \quad (11)$$

$$\psi^{n+1} = \psi^n + \omega (-\mu_0 R J - \Delta^* \psi) / a_C, \quad (12)$$

with R -dependent $a_{W,E} = 1/dR^2 \pm 1/(2RdR)$, $a_{N,S} = 1/dZ^2$, $a_C = -2(1/dR^2 + 1/dZ^2)$. Tests in-kernel reduction strategies (single-TG vs simdgroup-tree) and multi-kernel composition.

2.6 Data-shuffle / butterfly (regime R6)

fft3d. 3D complex-to-complex forward FFT, fp32, on a power-of-two cube of side N . Convention is unnormalised (matches `numpy.fft.fftn`); storage is row-major `float2[N][N][N]`. Three named kernels — `fft3d_x`, `fft3d_y`, `fft3d_z` — are dispatched in sequence with two ping-ponged buffers, each kernel performing a length- N 1D FFT per threadgroup of N cooperating threads:

$$Y[k_1, k_2, k_3] = \sum_{n_1, n_2, n_3=0}^{N-1} X[n_1, n_2, n_3] \exp\left(-\frac{2\pi i}{N}(k_1 n_1 + k_2 n_2 + k_3 n_3)\right). \quad (13)$$

Sizes $N \in \{32, 64, 128\}$ cover three working-set regimes: 32^3 (~ 256 KB) is SLC-resident and compute-bound; 128^3 (~ 16 MB) DRAM-binds. The optimisation surface is dominated by data movement, not arithmetic — bit-reversal vs Stockham auto-sort, twiddle-factor caching, mixed-radix (radix-4, radix-8) butterflies for fewer barriers, and `simd_shuffle` / `simd_shuffle_xor` for intra-simdgroup butterflies are all on the table. Tests threadgroup-memory bank-conflict avoidance and the per-axis stride asymmetry (stride 1 along x vs strides N and N^2 along y, z). $\sim 5N \log_2 N$ FLOPs per 1D FFT; effective DRAM traffic 96 B/cell across the three axis passes (16 B read + 16 B write per pass). Verification is max-norm against `numpy.fft.fftn` on a fixed seeded Gaussian input, tolerance $10^{-3} + 10^{-3} \|Y\|_\infty$.

A bandwidth-bound smoke-test (`saxpy`) is also included to validate the harness.

3 Harness

Compile and dispatch. The harness uses PyObjC’s Metal bindings and runtime-compiles `.metal` source via `MTLDevice.newLibraryWithSource`, avoiding the offline `xcrun metal` toolchain;

Table 1: Matched 10/15/25-iter sweeps on Apple M1 Pro: O = claude-opus-4-7, G = gemini-3.1-pro-preview. *In-dist.* $\times$ = best / seed, gmean over three training sizes; correctness is a hard gate. *Held-out frac.* = achieved/ceiling at the unseen size; *held-out* $\times$ = best / seed at that size. **Bold** marks regressions ($< 0.95\times$) and the HMC correctness fail.

Task	In-dist. $\times$		Held-out frac.		Held-out $\times$		Outcome
	O	G	O	G	O	G	
saxpy	1.25	1.00	78%	105%	0.79	1.06	saturated
heat2d	1.00	1.03	79%	84%	0.80	0.85	saturated
wave3d	1.26	1.00	99%	98%	1.02	1.01	saturated
ising	1.13	1.00	11%	12%	0.95	0.99	flat
fft3d	1.03	1.19	40%	45%	1.02	1.15	generalises
nbody	2.83	2.00	1.7%	1.6%	2.21	2.17	generalises
gradshaf	1.89	2.89	5.5%	8.6%	2.13	3.31	generalises
lj	1.77	1.98	0.25%	0.48%	1.16	2.19	generalises
lbm	1.46	1.06	44%	54%	0.41	0.50	both regress (TG geom.)
hmc	10.6	10.7	—	9.85%	FAIL	17.8	O wrong at $d=24$; G clean

compile errors are returned as structured strings to the LLM. Buffers are allocated in unified memory; numpy views alias buffer contents with a single copy on read-back. All dispatches for a multi-size run share one `MTLCommandBuffer`, and timings come from `GPUEndTime - GPUStartTime` (3 warmup, 10 timed, median reported). The chip is detected from `sysctl` and looked up in a per-family table (M1 through M4) for peak FP32 GFLOPS and DRAM bandwidth; all runs here are on Apple M1 Pro (4500 GFLOPS, 200 GB/s).

Scoring. For a candidate passing correctness at every size, $S = (\prod_i f_i)^{1/n}$ with $f_i = \text{achieved}_i / \text{ceiling}_i$. Any tolerance failure forces $S=0$; this hard gate prevents trading correctness for speed, and the geometric mean across sizes discourages overfitting to one regime.

Evolution loop. A single-population ($\mu=1+\lambda=1$) loop. Iteration i shows the LLM the previous candidate, the incumbent best, a short history per iteration, and the original task spec. The LLM returns a Metal source block (multi-kernel tasks emit multiple `kernel` functions in one file). Candidates strictly beating the incumbent are promoted; prompts, raw responses, extended-thinking tokens, sources, and per-size JSON results are persisted. A single `call_llm` bridge dispatches Claude via the Agent SDK and Gemini via `google-genai`.

4 Experiments

We run two matched single-model sweeps on Apple M1 Pro — claude-opus-4-7 (*O*) and gemini-3.1-pro-preview (*G*) — over the ten tasks at the same per-task iteration budget (10 each except lbm at 25 and wave3d at 15), $\mu=1+\lambda=1$, no human prompt intervention. Table 1 reports each model’s in-distribution self-speedup (best / seed, gmean over three training sizes) and a held-out evaluation in which the unmodified seed and each incumbent best are run on a single unseen problem size declared by the task spec (Sec. 2).

In-distribution. Self-speedups span $1.00\times$ to $10.7\times$. The HMC step (Fig. 1) is the high end *for both models*: each independently introduces a `template <uint D>` worker dispatched on runtime d that takes $d=8$ from ~ 120 to ~ 970 GFLOPS by enabling full unroll of the inner Aq matvec, and the two top scores agree to within 1.4% (O 0.0932 vs G 0.0870). Outside HMC the models split: Opus wins clearly on saxpy (1.25 vs 1.00), nbody (2.83 vs 2.00), lbm (1.46 vs 1.06), and wave3d (1.26 vs 1.00); Gemini wins on gradshaf (2.89 vs 1.89), fft3d (1.19 vs 1.03), and lj (1.98 vs 1.77). The pattern correlates with the type of optimisation lever: “tune the same algorithm tighter” (BW saturation, BGK fold, leapfrog ILP) favours Opus, while “find a different algorithm” (simdgroup-tree reduction, twiddle caching, neighbour-list reorganisation) favours Gemini (see App. A for code-level diffs on lbm and fft3d). Saturated tasks (saxpy, heat2d, wave3d) sit above 78% of effective DRAM ceiling on the seed; saxpy and heat2d primarily validate the harness, leaving the search loop little to do — and heat2d shows the score is a *hard* signal: on Opus zero candidates strictly dominated the seed across all three sizes and the incumbent stayed at iter 0. Gemini had *zero*

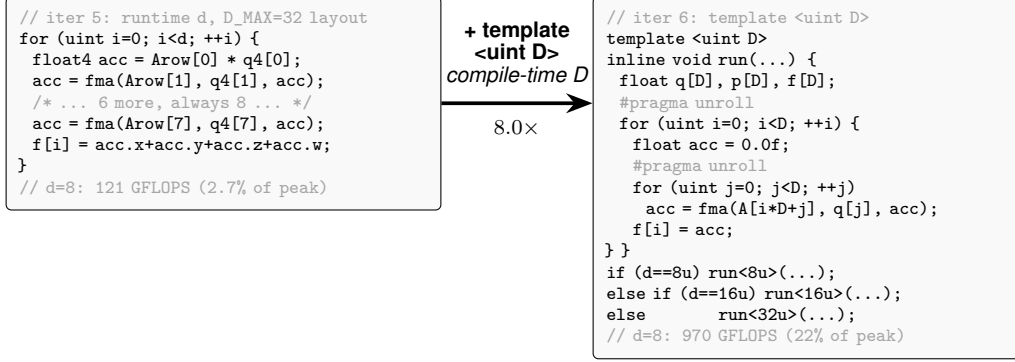


Figure 1: Paradigmatic candidate evolution: HMC, Opus 4.7, iter 5 $\rightarrow$ 6. Both Opus and Gemini independently arrive at this structural change. The lever is one declaration — `template <uint D>` with runtime-dispatched instantiations on d . Iter 5 had a manually-unrolled float4 inner loop against a fixed $D_{\max}=32$ layout (over-computing at $d=8$, plus a 4-way horizontal sum); iter 6 sizes q, p, f exactly to D so both loops fully unroll into scalar FMAs — moving $d=8$ from 121 to 970 GFLOPS in one iteration after five had stalled at 2–4% of peak. Opus enumerated $D \in \{8, 16, 32\}$ and broke correctness at the held-out $d=24$; Gemini kept a runtime- d fallback and generalised cleanly.

correctness failures across its 95 fresh-run candidates; Opus had 13 across the same candidate budget (notably wave3d 10/15: multi-step leapfrog amplifies any sign or indexing error into NaN).

Held-out generalisation is sharper than in-distribution — and the two models overfit asymmetrically. Three populations separate cleanly. (i) *Generalises*: nbody, fft3d, gradshaf, lj. Gradshaf is the only task where both models match or beat their in-distribution speedup ($2.13\times$ Opus, $3.31\times$ Gemini); on lj only Gemini extrapolates cleanly ($2.19\times$ at $N=2744$, above its in-distribution $1.98\times$), while Opus drops from $1.77\times$ to $1.16\times$. fft3d 256^3 extrapolates past the largest training size 128^3 for both models. (ii) *Saturated*: saxpy, heat2d, wave3d, ising — nothing to win. (iii) *Overfits*, with the most consequential disagreement between the two models. **HMC** is the sharpest case: Opus’s template specialisation dispatches `if (d==8) run<8>() ... else run<32>()`, so $d=24$ lands in the $D=32$ branch, per-thread $q[32]$, $p[32]$ and the unrolled matvec process 32 entries against 24-entry data, and sample covariance lands $\sim 10\sigma$ off target. Gemini arrived at the same template- D speedup in-distribution but kept a runtime- d leapfrog around it, which generalises to $d=24$ at 9.85% of FP32 peak — $17.8\times$ the seed. **LBM** regresses for both ($0.41\times$ Opus, $0.50\times$ Gemini) at the held-out 192^2 grid: each pins a fixed `[[max_total_threads_per_threadgroup]]` geometry tuned for the $\{64, 128, 256\}^2$ training sizes; on a non-multiple of 32 the geometry wastes threads at the boundary, and the BGK algebraic fold cannot recover the loss. **Gemini is never strictly worse than Opus on held-out across the ten tasks**: it is the only model correct at HMC $d=24$, posts a higher held-out speedup on six of the remaining nine tasks, and ties on the other three. The LBM 256^2 result above 100% of nominal ceiling reflects working-set residency in the system-level cache: the 8 B/cell roofline is DRAM-only.

Recurring Metal-grammar failures. Compile-error patterns are similar across models. `[[max_total_threads_per_threadgroup(N)]]` is mis-placed (after the parameter list, or as a standalone statement, instead of on the `kernel void` declaration) on Opus across five tasks — reproduced from earlier opus-4-6 runs. `half` is reserved as MSL’s fp16 type, breaking `uint half = N >> 1u;`; C++ lambdas are unsupported. The two sweeps differ in volume rather than kind: 12/110 compile fails for Opus vs 22/95 for Gemini, with Gemini exploring more aggressively per iteration but never breaking correctness, and Opus producing more correct-but-slow candidates. Gemini’s wall-time per iteration is 5–10 $\times$ Opus’s at matched budget, the cost of that wider exploration.

5 Discussion

The held-out gate as agent oversight. The benchmark’s (1+1) loop is, modulo terminology, an autonomous coding agent: it reads a Metal source, edits it, runs it, and self-promotes its own outputs

based on an internal fitness signal. A human merging the agent’s incumbent into a downstream code-base sees only the in-distribution score the agent reports — and that score is gameable in two ways our held-out gate catches. **Silent correctness violation.** On HMC, Opus’s incumbent hits $10.6\times$ with all in-distribution checks green; held out at $d=24$ — a problem dimension that sits between two training values and is not flagged as out-of-scope — the same code returns samples whose covariance is off by $\sim 10\sigma$. A user who trusted the reported number would ship a sampler that looks calibrated and isn’t. **Silent regression.** On LBM both models report in-distribution wins ($1.46\times$, $1.06\times$) that flip to slowdowns at the held-out grid ($0.41\times$, $0.50\times$); the agent confidently labels its output as “improvement” on a configuration only $1.5\times$ from the training sizes. Both failures share a structural shape: the agent specialised against the visible workload and reported a number that does not carry over. A single auxiliary problem instance per task — one extra dispatch, seconds of GPU time — is enough to surface both. This reframes the contribution toward the workshop’s verifiability/oversight agenda: METAL-SCI’s value lies less in providing a hard benchmark and more in instantiating a cheap, mechanical trust contract on top of an autonomous coding agent. The roofline anchor, per-size tolerance, and geometric mean are the inner scoring machinery; the held-out gate is the lightweight oversight primitive that lets a human (or a downstream automated reviewer) catch confidently-wrong agent code before deployment.

Other observations. A roofline-anchored score answers “how close are we to the hardware?” in physical units; the gmean across sizes is hard to game by overfitting one regime. The Opus–Gemini asymmetry is concrete: at matched iteration budgets the in-distribution scores are close, but Gemini’s optimisations preserve correctness across the held-out size on every task while Opus’s break it on one. If kernels will be deployed across configurations the training set did not cover, that robustness is worth the $5\text{--}10\times$ wall-time premium. Apple Silicon’s unified memory halves the host-side scaffolding compared to CUDA, enabling sub-second compile-run-verify cycles per candidate — which matters more than it sounds: an evolutionary loop with tens of candidates needs a fast *har-ness*, not a fast kernel. Metal’s smaller, quirkier surface also surfaces OOD failures of CUDA-trained LLMs on tasks that are otherwise textbook.

Limitations and future work. The static per-chip ceilings do not account for SLC residency at small sizes (see the LBM 256^2 result above); a workload-aware roofline [4] per task and per size would tighten the score. The single-population (1+1) loop plateaus within 10–25 iterations on most tasks — an island-model or FunSearch-style [1] archive with a novelty signal [2] is the natural next step, particularly for the irregular-memory tasks. Future tasks include sparse linear algebra (SpMV, CG) and cross-chip generalisation (evolve on M2 Pro, evaluate on M3 Max).

References

- [1] Romera-Paredes, B. et al. (2024) Mathematical discoveries from program search with large language models. *Nature* **625**, 468–475.
- [2] Novikov, A. et al. (2025) AlphaEvolve: A coding agent for scientific and algorithmic discovery. *arXiv:2506.13131*.
- [3] Ouyang, A. et al. (2025) KernelBench: Can LLMs Write Efficient GPU Kernels? *arXiv:2502.10517*.
- [4] Williams, S., Waterman, A. & Patterson, D. (2009) Roofline: An insightful visual performance model for multi-core architectures. *CACM* **52**(4), 65–76.
- [5] Datta, K. et al. (2008) Stencil computation optimization and auto-tuning on state-of-the-art multicore architectures. *Proc. SC '08*.
- [6] Nyland, L., Harris, M. & Prins, J. (2007) Fast N-body simulation with CUDA. *GPU Gems 3*, ch. 31.
- [7] Schönherr, M. et al. (2011) Multi-thread implementations of the lattice Boltzmann method on non-uniform grids for CPUs and GPUs. *Comput. Math. Appl.* **61**(12), 3730–3743.
- [8] Anderson, J. A., Lorenz, C. D. & Travesset, A. (2008) General purpose molecular dynamics simulations fully implemented on graphics processing units. *J. Comput. Phys.* **227**(10), 5342–5359.
- [9] Micikevicius, P. (2009) 3D finite difference computation on GPUs using CUDA. *Proc. GPGPU-2*.
- [10] Geb, L. et al. (2022) Seamless GPU acceleration for C++ based physics on the Apple M1’s unified processing units. *arXiv:2206.01791*.
- [11] Govindaraju, N. K. et al. (2008) High performance discrete Fourier transforms on graphics processors. *Proc. SC '08*.
- [12] Li, J. et al. (2025) TritonBench: Benchmarking large language model capabilities for generating Triton operators. *Findings of ACL 2025*, pp. 23053–23066.
- [13] Saroufim, M. et al. (2025) BackendBench: benchmarking PyTorch-backend kernel generation.
- [14] Wen, J. et al. (2025) MultiKernelBench: multi-platform LLM kernel generation across CUDA, Ascend C, and Pallas.

```

// Opus: BGK fold + pinned geometry
[[max_total_threads_per_threadgroup(64)]]
kernel void lbm_step(...) {
    // pull-stream f0..f8, moments rho, ux, uy ...
    float A = fma(-1.5f, usq, 1.0f);
    float orWS = (omega * W_S) * rho;
    // k=5: cu = ux+uy
    {
        float cu = ux + uy;
        float t = A + cu * fma(4.5f, cu, 3.0f);
        q5[idx] = fma(one_m_w, f5, orWS * t);
    }
    // 8 more directions, same shape ...
}

```

```

// Gemini: textbook BGK in unrolled loop
kernel void lbm_step(...) {
    // pull-stream f[k], moments rho, ux, uy ...
    float usq = ux*ux + uy*uy;
    float inv_tau = 1.0f / tau;
    #pragma unroll
    for (int k = 0; k < 9; ++k) {
        float cu = CX[k]*ux + CY[k]*uy;
        float feq = W[k] * rho *
            (1.0f + 3.0f*cu + 4.5f*cu*cu
             - 1.5f*usq);
        f_out[k*N+idx] =
            f[k] - inv_tau*(f[k]-feq);
    }
}

```

Figure 2: LBM iter-23 best, Opus (left) vs Gemini iter-13 best (right). Opus extracts A once, folds f_k^{eq} into two FMAs per direction and the relaxation into a third, and pins the threadgroup geometry; Gemini stays with the canonical BGK formula and the default geometry. In-distribution gmean: Opus 0.576 vs Gemini 0.553.

- [15] Kalade, S. & Schelle, G. (2025) NPUEval: optimising NPU kernels with LLMs and open-source compilers. *arXiv:2507.14403*.
- [16] Nie, J. et al. (2026) KernelCraft: benchmarking agentic close-to-metal kernel generation on emerging hardware. *arXiv:2603.08721*.
- [17] Lange, R. T. et al. (2025) Towards robust agentic CUDA kernel benchmarking, verification, and optimisation. *arXiv:2509.14279*.
- [18] Guo, P. et al. (2025) EvoEngineer: mastering automated CUDA kernel evolution with large language models. *arXiv:2510.03760*.

A Code-level evidence for the in-distribution split

The comparative claim in Section 4 is mechanistic — Opus wins by tightening the same algorithm, Gemini wins by reaching for a different one. Figure 1 instantiates that claim on HMC. This appendix instantiates it on two more tasks — one Opus-favoured (lbm), one Gemini-favoured (fft3d) — with the actual diff between each model’s incumbent best.

A.1 LBM: Opus tightens BGK with FMA folds and a pinned threadgroup

Both models share the seed’s pull-stream + BGK structure. The diff is local to the collision step and to the kernel attribute (Fig. 2). Opus precomputes $A = \text{fma}(-1.5, \|u\|^2, 1)$ once per cell, factors the per-direction equilibrium as $A + c \cdot u (3 + 4.5 c \cdot u)$ — two FMAs — and folds the relaxation into a third $\text{fma}(1 - \omega, f_k, \omega W_k \rho t)$, yielding nine manually unrolled blocks. It also pins `[[max_total_threads_per_threadgroup(64)]]` on the kernel declaration, picking a 32×2 tile aligned to the simdgroup width. Gemini keeps the textbook BGK formula $w_k \rho (1 + 3c \cdot u + 4.5(c \cdot u)^2 - 1.5\|u\|^2)$ inside a `#pragma unroll for (k=0...8)`, no FMA folds, no A -extraction, no threadgroup pin. The two are competitive at 64^2 (O 0.34, G 0.32) and Gemini actually wins at 128^2 (O 0.47, G 0.51); Opus pulls ahead at 256^2 (O 1.22, G 1.03), the cache-resident regime where the pinned 32×2 geometry and the FMA-folded BGK extract every issued instruction — and that is the size with the largest absolute fraction-of-ceiling, so it dominates the gmean.

A.2 FFT3D: Gemini swaps the algorithm to `simd_shuffle_xor`

The fft3d gmean gap is larger (0.282 vs 0.167, $1.7\times$), and the diff is not a tighter version of the same kernel — Fig. 3 shows two different algorithms. Opus implements a textbook Stockham auto-sort radix-4 FFT: every stage ping-pongs through threadgroup memory, with a barrier between stages. Gemini observes that for the first five Cooley–Tukey stages the butterfly partner of lane i is at lane $i \oplus 2^{s-1}$ where $s \in \{1, \dots, 5\}$, and $2^{s-1} < 32$ — so it can be fetched with `simd_shuffle_xor`, no shared memory and no barrier. Only stages $s \geq 6$ fall back to threadgroup memory. The Apple GPU’s simdgroup width is exactly 32: this trick is Metal-specific (`simd_shuffle_xor` maps to a single hardware permute), and worth ~ 5 barriers per length- N FFT, ~ 15 per 3D transform. The same “find a different memory primitive” move shows up smaller-scale on gradshaf: both models converge on a simdgroup-tree max-reduction, but Gemini additionally casts `psi` to `float4*` and reads four scalars per transaction in the inner sweep, halving the number of issued loads on the dominant kernel.

B Related work

B.1 LLM-driven kernel-generation benchmarks

A wave of benchmarks has emerged for evaluating LLMs as generators of high-performance GPU kernels. *KernelBench* [3] targets PyTorch ML kernels (GEMM, attention, convolution, normalisation, activation) on CUDA and scores single-shot

```

// Opus: Stockham radix-4, all in tg memory
for (uint s = 0u; s < stages4; ++s) {
    // load 4 inputs from cur[]
    float2 x0 = cur[b + 0u*Nq];
    float2 x1 = cur[b + 1u*Nq];
    float2 x2 = cur[b + 2u*Nq];
    float2 x3 = cur[b + 3u*Nq];
    // twiddle multiplies (skip s=0)
    if (s != 0u) { ... }
    // 4-point DFT, write to nxt[]
    threadgroup_barrier(mem_threadgroup);
    swap(cur, nxt);
}

```

```

// Gemini: simd_shuffle_xor for stages 1..5
// (no shared mem, no barrier)
if (logN >= 1u) {
    float2 v = simd_shuffle_xor(u, 1u);
    u = ((i & 1u) == 0u) ? (u+v) : (v-u);
}
// stages 2..4 analogous (xor 2, 4, 8)
if (logN >= 5u) {
    float2 v = simd_shuffle_xor(u, 16u);
    // twiddle, butterfly ...
    u = ((i & 16u) == 0u) ? (u+t) : (v-t);
}
// stages s>=6: fall back to tg memory
for (uint s = 6u; s <= logN; ++s) { ... }

```

Figure 3: FFT3D iter-10 best, Opus (left) vs Gemini iter-10 best (right). Opus implements Stockham radix-4 with threadgroup-memory ping-pong and a barrier per stage; Gemini exploits the 32-wide simdgroup to do the first five stages with `simd_shuffle_xor`, eliminating five barriers per 1D FFT. In-distribution gmean: Opus 0.167 vs Gemini 0.282.

generation by speedup against the PyTorch eager baseline. *TritonBench* [12] extends to the Triton DSL and adds reference code-similarity to correctness and performance. *BackendBench* [13] evaluates correctness alone for kernels written against the PyTorch backend interface. *MultiKernelBench* [14] broadens the platform set to CUDA, Ascend C, and Pallas while retaining the ML-operator workload set. *NPUEval* [15] targets the AMD AIE NPU in C++ and exposes a tool-use feedback loop with multi-turn regeneration. Most recently, *KernelCraft* [16] benchmarks bare-metal assembly kernel generation on emerging accelerator ISAs (PLENA, AMD NPU, Coral NPU, Sonic BOOM), with native tool interfaces (`check_syntax`, `run_evaluation`, `view_output`, `grep_docs`) and a three-tier ML-task taxonomy (primitive, composite, end-to-end). Across these benchmarks the workload is ML and the headline number is speedup against a vendor or compiler baseline.

B.2 LLM-driven evolutionary code search

The broader paradigm of LLMs as *optimisers* inside evolutionary code-search loops was established by *FunSearch* [1] for symbolic-algorithm discovery and scaled by *AlphaEvolve* [2] across mathematical and algorithmic domains. *AI CUDA Engineer* [17] and *EvoEngineer* [18] specialise this paradigm to CUDA kernel optimisation, replacing single-shot or agentic regeneration with a population/archive driven by a fitness signal — closer in spirit to our ($\mu=1+\lambda=1$) loop, but on CUDA ML kernels rather than Apple Silicon scientific kernels.

B.3 Where METAL-SCI differs

Existing kernel-generation benchmarks share three structural choices that METAL-SCI deviates from. (i) *Workload domain*. All listed benchmarks target ML operators, where the optimisation surface is dominated by tiling and warp-level reduction patterns heavily represented in pretraining data. METAL-SCI targets scientific compute — regular and irregular stencils, all-pairs interactions, multi-field Boltzmann updates, neighbour-list molecular dynamics, parallel-chain MCMC, multi-kernel nonlinear PDE solvers, and butterfly spectral transforms — six structurally distinct optimisation regimes (R1–R6 of Section 2) whose canonical recipes [5,6,7,8,11] do not transfer between regimes. (ii) *Score basis*. All listed benchmarks normalise *achieved* against a vendor or compiler baseline (PyTorch eager, compiler-emitted code, expert hand-tuned reference). This couples the headline number to the strength of that baseline. METAL-SCI normalises against the per-chip roofline ceiling (peak FP32 GFLOPS, peak DRAM GB/s), a hardware-anchored target independent of any reference implementation, with a hard correctness gate ($S=0$ on tolerance failure) and a geometric mean across three problem sizes that discourages overfit to a single working-set regime. (iii) *Hardware*. No listed benchmark targets Apple Silicon’s Metal compute pipeline. Metal is structurally underrepresented in CUDA-centric pretraining data and surfaces a real out-of-distribution generalisation test on Metal-grammar and Metal-stdlib idioms — the `[[max_total_threads_per_threadgroup]]` attribute placement (Sec. 4, seven independent reproductions across five tasks within the opus-4-7 sweep, recurring in earlier opus-4-6 and gemini-3.1-pro runs), `half` as a reserved fp16 keyword, the absence of C++ lambdas, the availability of `simd_shuffle_xor` for intra-simdgroup butterflies, and the absence of `sincospi`. Table 2 summarises these axes against the comparator set.

Table 2: METAL-SCI versus existing LLM kernel-generation benchmarks across the axes that drive the design of our harness. *Domain*: ML = neural-network operators (GEMM, attention, convolution, normalisation, activation); HPC = scientific compute (stencil, N -body, LBM, MD, MCMC, PDE solver, FFT). *Score basis*: what *achieved* is normalised against. *Multi-size*: whether the score requires generalisation across problem sizes. *Search*: the loop structure exposed to the LLM. Within the ML row, all listed benchmarks span a single optimisation regime (matmul-style, with norm/activation epilogues); METAL-SCI spans six structurally distinct regimes (R1–R6 of Section 2).

Benchmark	Target	Domain	Score basis	Multi-size	Search
KernelBench [3]	CUDA	ML	PyTorch speedup	—	single-shot
TritonBench [12]	Triton DSL	ML	speedup + code-sim	—	single-shot
BackendBench [13]	PyTorch backend	ML	correctness only	—	single-shot
MultiKernelBench [14]	CUDA / Ascend C / Pallas	ML	compiler speedup	—	multi-turn
NPUEval [15]	AIE C++ (AMD NPU)	ML	compiler speedup	—	tool-use
KernelCraft [16]	accelerator asm	ML	compiler speedup	cfg. vars	agentic tool-use
METAL-SCI (ours)	Apple Metal MSL	HPC	% of roofline	3 in + 1 held-out	evolutionary (1+1)